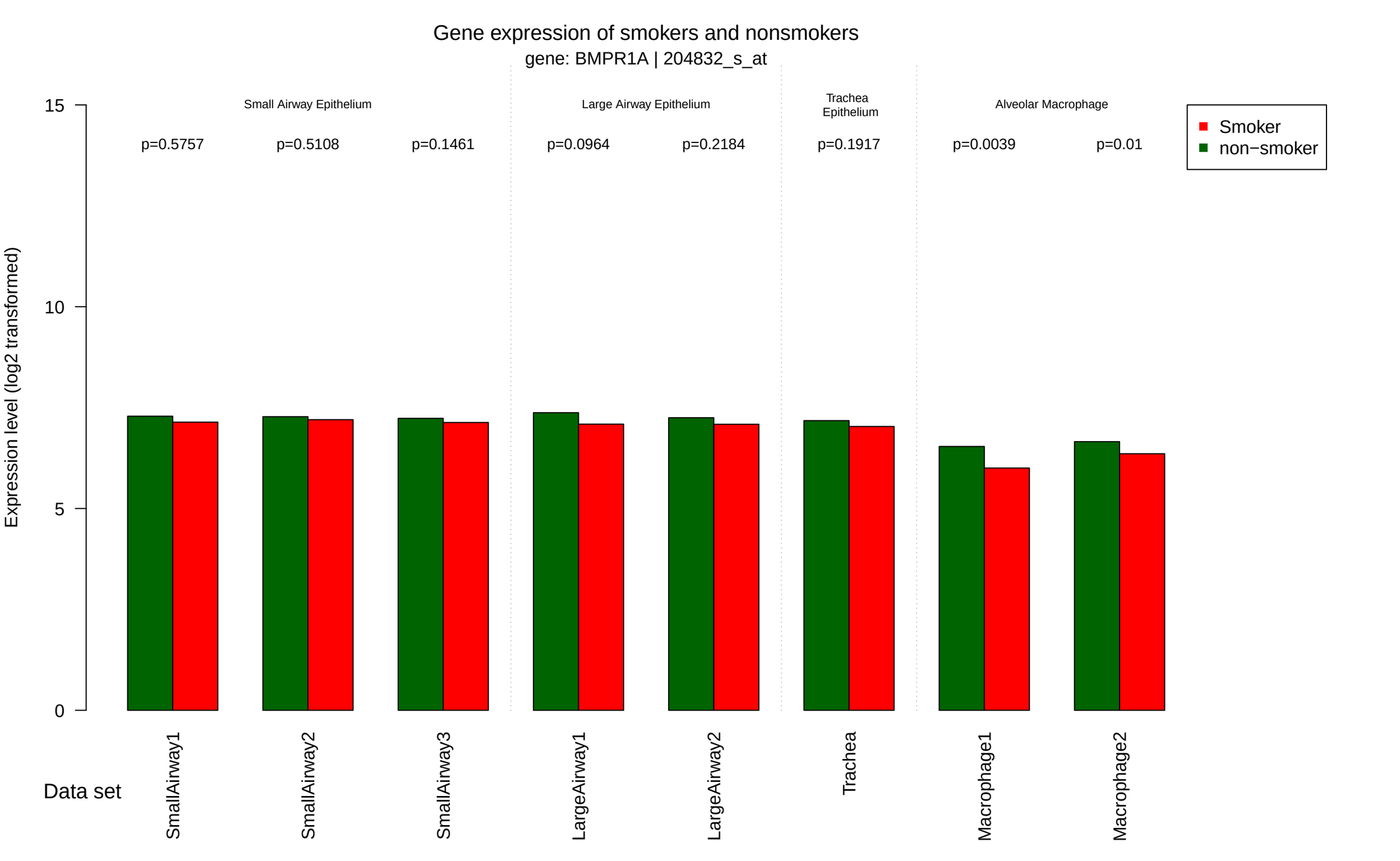
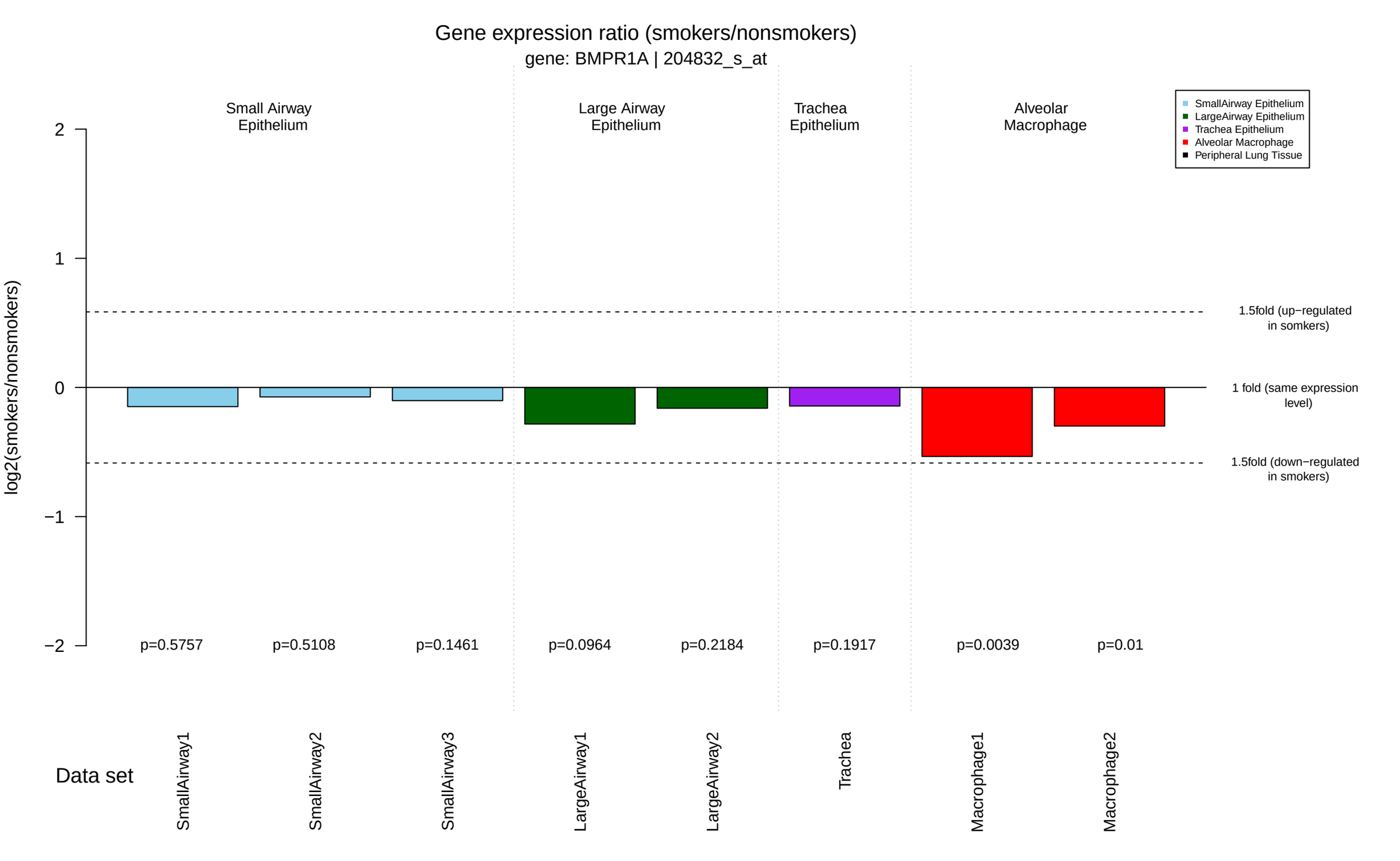
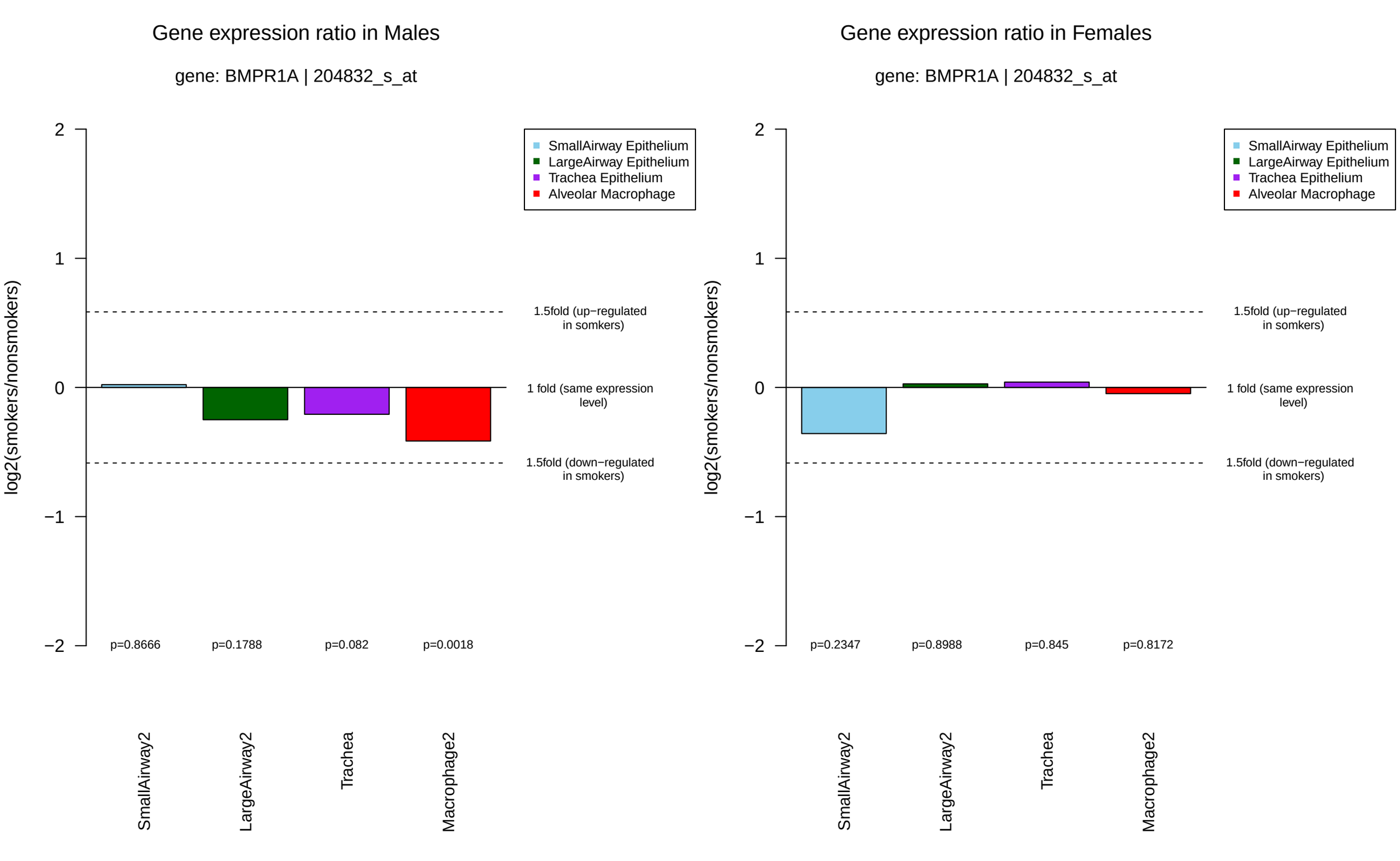
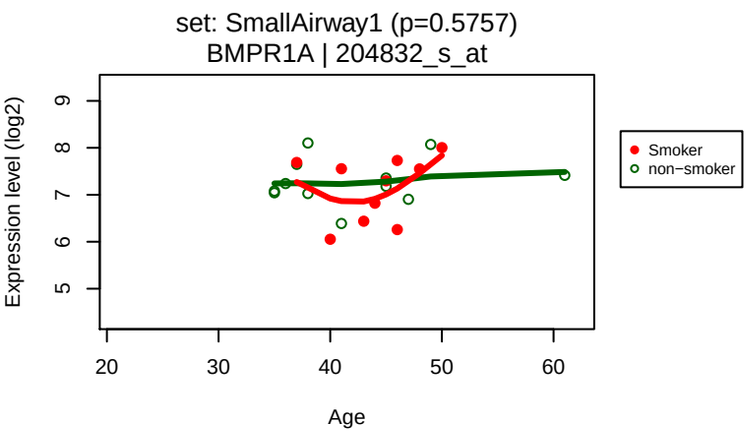
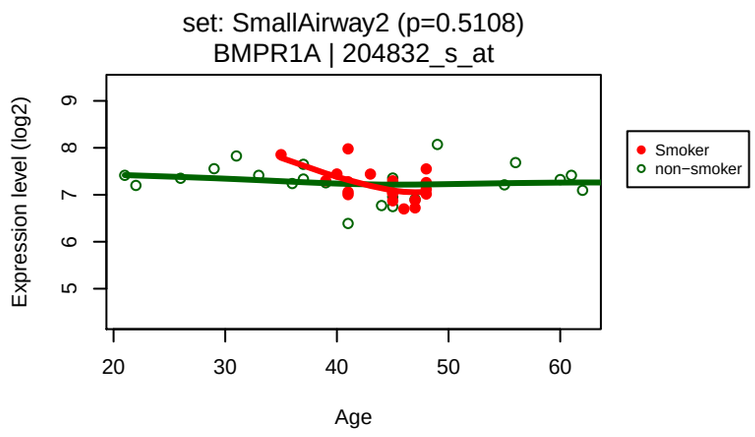
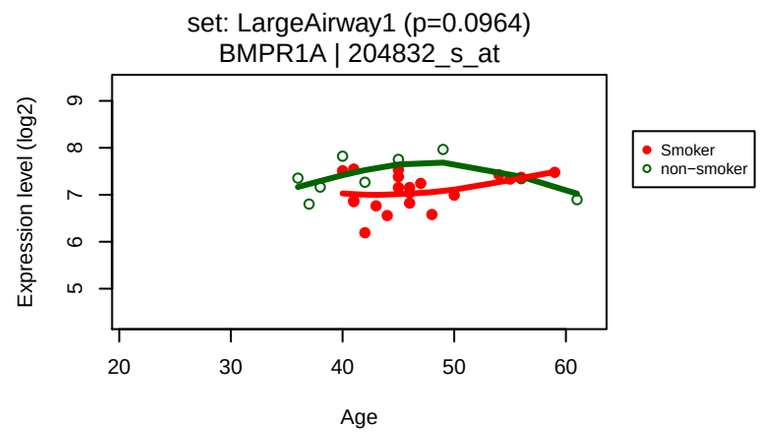

Gene expression of smokers and nonsmokers
gene: BMPR1A | 204832_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5757
p=0.5108
p=0.1461
p=0.0964
p=0.2184
p=0.1917
p=0.0039
p=0.01
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

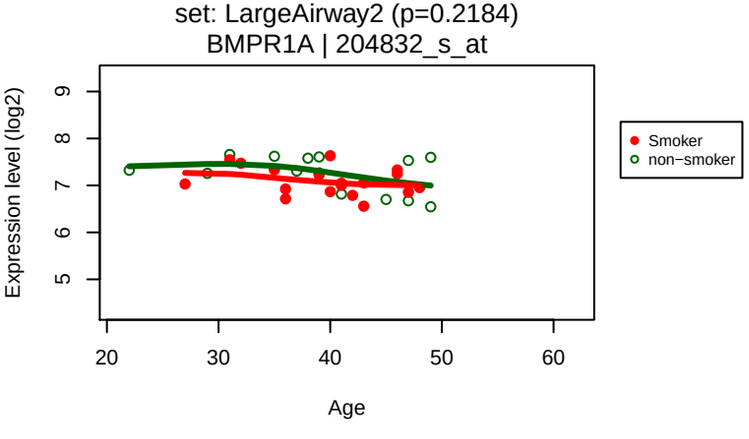

Gene expression ratio (smokers/nonsmokers)
gene: BMPR1A | 204832_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5757
p=0.5108
p=0.1461
p=0.0964
p=0.2184
p=0.1917
p=0.0039
p=0.01
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

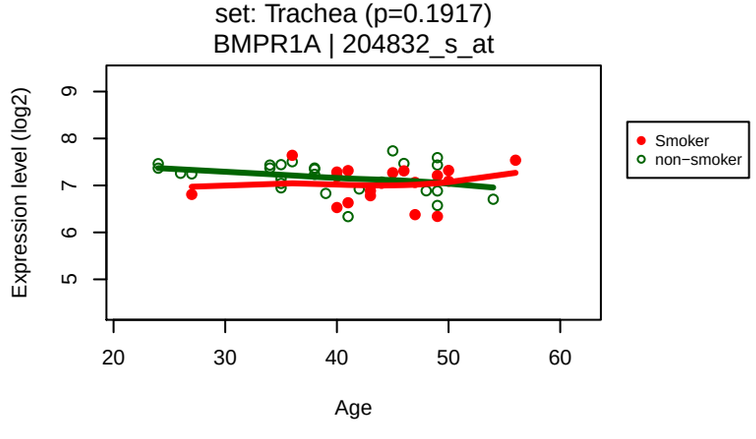

Gene expression ratio in Males
gene: BMPR1A | 204832_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8666
p=0.1788
p=0.082
p=0.0018
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BMPR1A | 204832_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2347
p=0.8988
p=0.845
p=0.8172
SmallAirway2
LargeAirway2
Trachea
Macrophage2

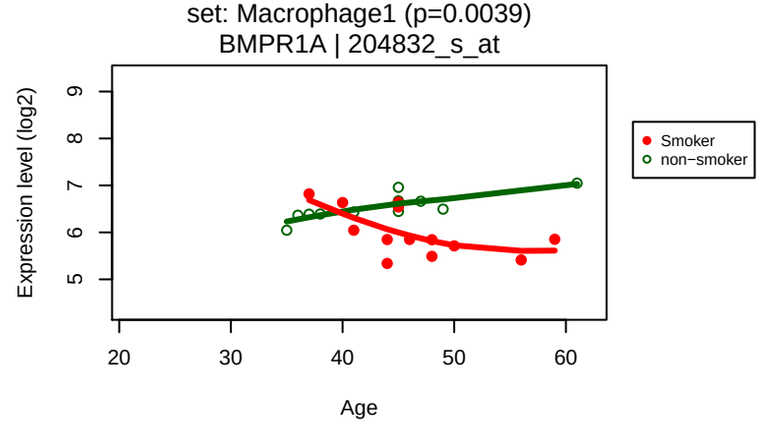

set: SmallAirway1 (p=0.5757)
BMPR1A | 204832_s_at
Expression level (log2)
Age
Smoker
non−smoker

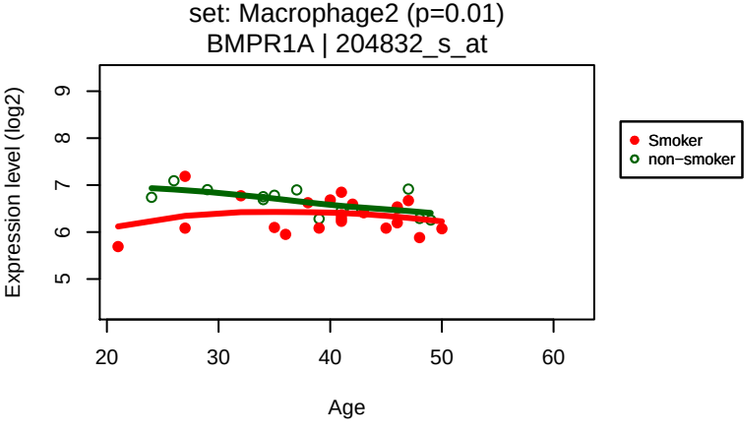

set: SmallAirway2 (p=0.5108)
BMPR1A | 204832_s_at
Expression level (log2)
Age
Smoker
non−smoker

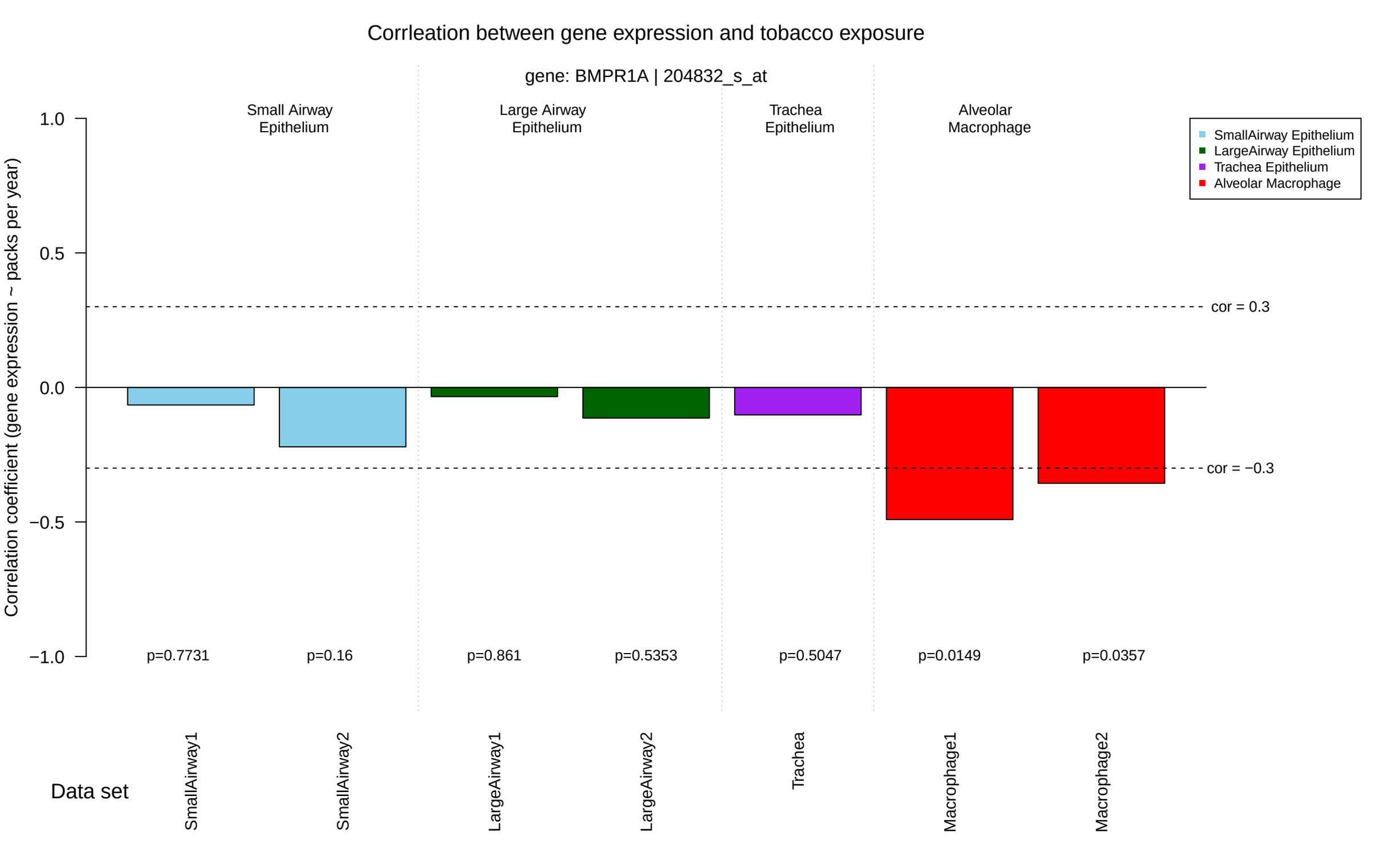

set: LargeAirway1 (p=0.0964)
BMPR1A | 204832_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2184)
BMPR1A | 204832_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1917)
BMPR1A | 204832_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0039)
BMPR1A | 204832_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.01)
BMPR1A | 204832_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BMPR1A | 204832_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7731
p=0.16
p=0.861
p=0.5353
p=0.5047
p=0.0149
p=0.0357
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2